Weighted Synthetic Difference-in-Differences

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Abstract

Replacing the equal treated-unit weighting of synthetic difference-in-differences (SDID) with population weights moves the estimated county-level mortality effect of the 2014 ACA Medicaid expansion from a precise null to a reduction of roughly 17 deaths per 100,000. The divergence is driven entirely by the choice of estimand, not of estimator, and reflects treatment effects concentrated in the largest urban counties. We extend SDID to researcher-specified treated-unit weights through a single modification of the collapsed form, adapt the bootstrap, jackknife, and placebo variance estimators with cluster-robust versions, and establish consistency and asymptotic normality under a condition requiring the effective number of treated units to grow. Because that condition is strained in many applications, including ours, we recommend cluster-robust bootstrap inference, and we propose a paired-bootstrap test of the gap between estimands as an ex-ante practitioner diagnostic. The estimator extends the `synthdid` R package.

Keywords: Synthetic difference-in-differences, causal inference, panel data, heterogeneous treatment effects, Medicaid expansion

JEL Codes: C21, C23, I13

1 Introduction

The synthetic difference-in-differences (SDID) estimator of Arkhangelsky et al. (2021) has become a workhorse for causal inference in panel settings, combining the strengths of synthetic control (Abadie et al., 2010) and difference-in-differences by using data-driven unit weights $\hat{\omega}$ and time weights $\hat{\lambda}$ while permitting level differences across units. When multiple units are treated, the standard SDID implementation averages them equally: each treated unit contributes $1/N_1$ to both the target estimand and the construction of the synthetic control. This is an implicit design choice, not a requirement of the methodology.

The choice has consequences. When treatment effects are heterogeneous and correlated with unit size, the equally-weighted estimand can diverge sharply from estimands that researchers actually

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care about—per-capita treatment effects, average effects on the population exposed to treatment, or effects on subpopulations of policy interest. A growing literature has emphasized that aggregation choices in panel estimators are not innocuous: Gibbons et al. (2019) show that two-way fixed-effects regressions implicitly weight observations in ways that may diverge from any natural causal estimand; Słoczyński (2022) extends this critique to OLS with heterogeneous effects; Callaway and Sant'Anna (2021) and de Chaisemartin and D'Haultfoeuille (2020) develop frameworks for staggered-adoption settings in which conventional aggregations can produce negatively-weighted combinations of unit-specific effects. The synthetic control literature has similarly seen extensions designed to address aggregation and identification concerns: Ben-Michael et al. (2021) propose augmented synthetic control with covariate adjustment, Cattaneo et al. (2021) develop prediction-interval inference, and Pang et al. (2022) propose a Bayesian alternative. SDID has so far been left out of this conversation, despite its rapid empirical adoption.

We make three contributions. First, we develop a weighted SDID estimator that replaces the uniform $1/N_1$ averaging with researcher-chosen weights $\tilde{\omega}_j$ in the collapsed form of the outcome matrix. The modification is parsimonious—a single change to the optimization target—and propagates naturally through the data-driven control and time weights $\hat{\omega}^w, \hat{\lambda}^w$. Standard SDID is recovered when $\tilde{\omega}_j = 1/N_1$, and weighted DID and weighted SC emerge as boundary cases. Second, we establish consistency and asymptotic normality of the weighted estimator under conditions that inherit from the original SDID factor-model framework, supplemented by a Lindeberg-type requirement that the *effective* number of treated units—the Kish $N_1^{\text{eff}} = (\sum_j \tilde{\omega}_j^2)^{-1}$—grow with N_1. We document that this condition loses its practical bite in many policy-relevant applications, including ours, and motivate cluster-robust bootstrap inference as the appropriate finite-sample tool. Third, we adapt all three variance estimators of Arkhangelsky et al. (2021) (bootstrap, jackknife, and placebo) to the weighted case with appropriate weight renormalization, and extend the bootstrap to handle clustered treatment assignment following Clarke et al. (2023). The full set of estimators is implemented as an extension to the `synthdid` R package.[1]

We apply the method to the county-level mortality effects of the ACA's 2014 Medicaid expansion, following the design of Borgschulte and Vogler (2020). They use double-lasso propensity score (DLPS) weighting to address pre-existing differences in mortality trends between expansion and non-expansion counties—a setting where SDID's data-driven reweighting is a natural alternative. Treated counties range from under 10,000 to over 10 million residents, making the weighting choice unambiguously consequential. The equally-weighted (i.e., standard) SDID estimate is statistically

[1]Available at https://github.com/johniselin-econ/synthdid_weights. Replication code and a companion supplementary appendix with proofs and additional robustness checks are also provided.

indistinguishable from zero. The population-weighted SDID estimate is approximately -17.5 deaths per 100,000, significant at the 1% level and of comparable magnitude to the -11.36 that Borgschulte and Vogler (2020) obtain via DLPS weighting. The divergence from the equally-weighted estimate is driven entirely by the weighting choice, not the estimator: a parallel weighted DID exercise produces qualitatively identical results. A binned scatter of unit-level treatment effects against log county population shows that effects in small and mid-sized counties oscillate around zero while the largest urban counties exhibit reductions exceeding 25 deaths per 100,000—the heterogeneity-size gradient that drives the result. The estimate is robust to alternative population-weight specifications (2010 vs. 2013 base years, square-root population), to leave-one-out exclusion of large counties, and to standard placebo tests. It is *not* robust to compressing the weight distribution toward equality (log population shares are sufficient to eliminate the effect), confirming that the divergence between estimands is real and quantitatively meaningful.

Section 2 introduces the weighted estimator, develops its asymptotic properties, and discusses inference; Section 3 applies the method to the ACA Medicaid expansion; Section 4 reports Monte Carlo evidence on finite-sample bias and confidence-interval coverage; Section 5 concludes.

2 The Weighted SDID Estimator

2.1 Setup

Consider a balanced panel with N units observed over T periods. The first N_0 units are untreated controls; the remaining $N_1 = N - N_0$ are treated beginning at period $T_0 + 1$, with $T_1 = T - T_0$ post-treatment periods. Define the treatment indicator $W_{it} = \mathbf{1}\{i > N_0\} \cdot \mathbf{1}\{t > T_0\}$ and observed outcomes $Y_{it} = Y_{it}(0) + \tau_{it} W_{it}$, where τ_{it} is the unit-time treatment effect.

Following Arkhangelsky et al. (2021), the standard SDID estimator solves

$$(\hat{\tau}^{\text{sdid}}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \underset{\tau,\mu,\alpha_i,\beta_t}{\arg\min} \sum_{i=1}^{N} \sum_{t=1}^{T} (Y_{it} - \mu - \alpha_i - \beta_t - \tau \cdot W_{it})^2 \, \hat{\omega}_i \hat{\lambda}_t, \tag{1}$$

where $\hat{\omega}_i$ and $\hat{\lambda}_t$ are data-driven unit and time weights. This admits a closed-form solution with a *collapsed form* $\tilde{Y}$, an $(N_0 + 1) \times (T_0 + 1)$ matrix in which the last row averages treated units and the last column averages post-treatment periods:

$$\tilde{Y}_{N_0+1,t} = \frac{1}{N_1} \sum_{j=1}^{N_1} Y_{N_0+j,t}, \quad \tilde{Y}_{i,T_0+1} = \frac{1}{T_1} \sum_{t=T_0+1}^{T} Y_{it}. \tag{2}$$

Control unit weights $\hat{\omega} \in \Delta^{N_0}$ and time weights $\hat{\lambda} \in \Delta^{T_0}$ solve the penalized matching programs

$$(\hat{\omega}_0, \hat{\omega}) = \underset{\omega_0 \in \mathbb{R},\, \omega \in \Delta^{N_0}}{\arg\min} \sum_{t=1}^{T_0} \Big(\omega_0 + \sum_{i=1}^{N_0} \omega_i Y_{it} - \tilde{Y}_{N_0+1,t}\Big)^2 + \zeta^2 T_0 \|\omega\|_2^2, \tag{3}$$

$$(\hat{\lambda}_0, \hat{\lambda}) = \underset{\lambda_0 \in \mathbb{R},\, \lambda \in \Delta^{T_0}}{\arg\min} \sum_{i=1}^{N_0} \Big(\lambda_0 + \sum_{s=1}^{T_0} \lambda_s Y_{is} - \tilde{Y}_{i,T_0+1}\Big)^2, \tag{4}$$

where ζ is the regularization level of Arkhangelsky et al. (2021): the intercepts permit level differences between the treated average and its synthetic control, and the ridge penalty spreads $\hat{\omega}$ across control units. Both programs are solved by Frank-Wolfe iterations on the simplex. The treatment effect is the bilinear form

$$\hat{\tau}^{\text{sdid}} = \begin{pmatrix} -\hat{\omega} \\ \frac{1}{N_1}\mathbf{1}_{N_1} \end{pmatrix}^{\top} Y \begin{pmatrix} -\hat{\lambda} \\ \frac{1}{T_1}\mathbf{1}_{T_1} \end{pmatrix}. \tag{5}$$

The key observation that motivates our extension is that the equal weights $1/N_1$ appear in two distinct places: in the collapsed form (2) that determines *which* synthetic control is constructed, and in the treated-row block of (5) that determines *which* treated average is being matched. In standard SDID, both are uniform by convention. Researchers who care about a non-uniform target estimand—per-capita effects, effects weighted by exposure or population—should reweight both.

2.2 The Weighted Modification

Let $\tilde{\omega} = (\tilde{\omega}_1, \ldots, \tilde{\omega}_{N_1})$ be researcher-specified non-negative weights on treated units, normalized so $\sum_j \tilde{\omega}_j = 1$. The target estimand is

$$\tau^{(\tilde{\omega})} = \sum_{j=1}^{N_1} \tilde{\omega}_j \, \bar{\tau}_j, \quad \bar{\tau}_j = T_1^{-1} \sum_{t=T_0+1}^{T} \tau_{N_0+j,t}. \tag{6}$$

We modify SDID by replacing the uniform average in the treated row of the collapsed form with the weighted average:

$$\tilde{Y}^w_{N_0+1,t} = \sum_{j=1}^{N_1} \tilde{\omega}_j Y_{N_0+j,t}, \quad \tilde{Y}^w_{i,T_0+1} = \frac{1}{T_1} \sum_{t=T_0+1}^{T} Y_{it}. \tag{7}$$

Control weights $\hat{\omega}^w$ and time weights $\hat{\lambda}^w$ are estimated from $\tilde{Y}^w$ via the same Frank-Wolfe algorithm (no modification to the solver is needed) and the weighted treatment effect is

$$\hat{\tau}^{w} = \begin{pmatrix} -\hat{\omega}^w \\ \tilde{\omega} \end{pmatrix}^{\top} Y \begin{pmatrix} -\hat{\lambda}^w \\ \frac{1}{T_1}\mathbf{1}_{T_1} \end{pmatrix}. \tag{8}$$

The modification is concentrated in two places: the collapsed form that feeds the weight solver, and the treated-unit block of the bilinear form. Equation (8) preserves the double-difference structure and delivers two boundary cases:

- **Standard SDID:** Setting $\tilde{\omega}_j = 1/N_1$ for all j recovers $\hat{\tau}^{\text{sdid}}$ exactly. (Proof: (7) reduces to (2), the Frank-Wolfe solutions coincide, and (8) reduces to (5).)
- **Weighted DID and weighted SC:** Setting $\hat{\omega} = \mathbf{1}_{N_0}/N_0$ and $\hat{\lambda} = \mathbf{1}_{T_0}/T_0$ produces a weighted DID estimator; setting $\hat{\lambda} = \mathbf{0}$ produces a weighted synthetic control estimator.

The interpretation of (8) as a sample analogue of (6) is that the synthetic control trajectory is constructed to match the weighted treated trajectory in pre-treatment periods, so that the post-treatment double-difference recovers the weighted average of unit-level treatment effects.

A two-unit example. The choice of $\tilde{\omega}$ matters exactly when treatment effects covary with the weights. From (6), the gap between the weighted and equally-weighted estimands is

$$\tau^{(\tilde{\omega})} - \tau^{(1/N_1)} = \sum_{j=1}^{N_1} \left(\tilde{\omega}_j - \frac{1}{N_1} \right) \bar{\tau}_j = \sum_{j=1}^{N_1} \left(\tilde{\omega}_j - \frac{1}{N_1} \right) (\bar{\tau}_j - \bar{\bar{\tau}}\,) \,,$$

where $\bar{\bar{\tau}} = N_1^{-1} \sum_j \bar{\tau}_j$ (the second equality holds because the weight deviations sum to zero)—an empirical covariance between weights and unit-level effects. Concretely, let $N_1 = 2$ with one large treated unit ($\tilde{\omega}_1 = 0.9$, $\bar{\tau}_1 = -15$) and one small unit ($\tilde{\omega}_2 = 0.1$, $\bar{\tau}_2 = +10$). The equally-weighted estimand is $(-15 + 10)/2 = -2.5$; the weighted estimand is $0.9(-15) + 0.1(10) = -12.5$. Under the assumptions of Section 2.3, each estimator is consistent for its own target, so the gap is not a bias of either: the two numbers answer different questions. Heterogeneity alone is not sufficient—reversing the weights ($\tilde{\omega}_1 = 0.1$) moves the weighted estimand to $+7.5$, and weights uncorrelated with effects leave the two estimands equal. Both heterogeneous effects *and* weight–effect correlation are required, which is what the binned scatter of Section 3.3 diagnoses in our application.

2.3 Asymptotic Theory

We establish consistency and asymptotic normality of $\hat{\tau}^w$ under conditions that inherit from the factor-model framework of Arkhangelsky et al. (2021), supplemented by regularity conditions on the treated-unit weights. The data-generating-process assumptions, stated formally as Assumptions A1–A5 in the supplementary appendix, are: (A1) outcomes follow a low-rank-plus-noise structure

$Y_{it} = L_{it} + \tau_{it} W_{it} + \epsilon_{it}$ with $L_{it} = \mu_{it} + \alpha_i + \beta_t$ for some low-rank μ; (A2) sub-Gaussian errors, independent across units and weakly dependent over time; (A3) factor-structure regularity; (A4) sample sizes $N_0, T_0 \to \infty$ with $\log N_0 = o(T_0)$ and $\log T_0 = o(N_0)$; (A5) existence of oracle weights (ω^*, λ^*) on the intercept-augmented simplex that match the weighted treated factor structure. We also impose two conditions on the treated-unit weights:

(W1) $\tilde{\omega}_j \geq 0$ and $\sum_j \tilde{\omega}_j = 1$.

(W2) $\max_j \tilde{\omega}_j \to 0$ as $N_1 \to \infty$, equivalently, the Kish effective sample size $N_1^{\text{eff}} := \left(\sum_j \tilde{\omega}_j^2\right)^{-1} \to \infty$.

Condition (W2) rules out a single dominant treated unit and is the key requirement for the central-limit machinery to apply to the weighted treated average. It is automatically satisfied by uniform weights ($N_1^{\text{eff}} = N_1$). For population weights, it is satisfied when treated units' size shares decay at any rate (e.g., a Pareto distribution with finite second moment), but it can fail when one or two units carry a non-vanishing share of total weight in the limit. We return to this point below.

Proposition 1 (Consistency). *Under Assumptions A1–A5 of Arkhangelsky et al. (2021) and (W1)–(W2),*

$$\hat{\tau}^w \xrightarrow{p} \tau^{(\tilde{\omega})} \quad \text{as } N_0, N_1, T_0 \to \infty.$$

Proof sketch. Following the decomposition argument of Arkhangelsky et al. (2021), write

$$\hat{\tau}^w - \tau^{(\tilde{\omega})} = \underbrace{\sum_j \tilde{\omega}_j \xi_j}_{\text{treated noise (i)}} + \underbrace{\Delta_{\text{ctrl}}}_{\text{control noise (ii)}} + \underbrace{\Delta_{\text{est}}}_{\text{weight estimation (iii)}},$$

where ξ_j is the post-minus-pre noise *contrast* for treated unit j—its post-period noise average minus the oracle-λ^*-weighted pre-period noise sum—with bounded variance under (A2); the contrast form absorbs within-unit serial covariance between post- and pre-period noise. For (i): $\text{Var}\left(\sum_j \tilde{\omega}_j \xi_j\right) \leq C \sum_j \tilde{\omega}_j^2 = C/N_1^{\text{eff}} \to 0$ by (W2), so (i) is $o_p(1)$ by Markov's inequality. For (ii): the analogous contrast at the oracle control weights has variance bounded by $C\|\omega^*\|_2^2 \to 0$ under (A5). For (iii): the weighted target column of the collapsed form remains a fixed convex combination of treated trajectories with an oracle representation in the control block (A5), so the weight-estimation error bounds of Arkhangelsky et al. (2021) apply with $\tilde{\omega}$ in place of $1/N_1$, and the surviving cross term—a product of loading-space estimation errors—vanishes under (A3)–(A4). Combining (i)–(iii) gives consistency. □

Proposition 2 (Asymptotic normality). *Under A1–A5, (W1)–(W2), the Lindeberg condition (W3):* $\max_j \tilde{\omega}_j^2 / \sum_k \tilde{\omega}_k^2 \to 0$, *and a negligibility condition (R) requiring the control-noise and weight-estimation terms to vanish at the* $\sqrt{N_1^{\mathit{eff}}}$ *scale,*

$$\sqrt{N_1^{\text{eff}}} \cdot \left(\hat{\tau}^w - \tau^{(\tilde{\omega})}\right) \xrightarrow{d} \mathcal{N}\left(0, \sigma_\xi^2\right),$$

where σ_ξ^2 *is the limiting variance of the standardized weighted treated-noise contrast.*

Proof sketch. Component (i) of the decomposition above is a Lindeberg-CLT-eligible weighted sum: $\sqrt{N_1^{\text{eff}}} \sum_j \tilde{\omega}_j \xi_j$ converges in distribution to $\mathcal{N}(0, \sigma_\xi^2)$ provided no single weight dominates the variance contribution—which is precisely condition (W3). Because ξ_j is a post-minus-pre contrast, σ_ξ^2 absorbs the within-unit serial covariance between post- and pre-period noise rather than assuming it away. Components (ii) and (iii) are $o_p(1)$ at the $\sqrt{N_1^{\text{eff}}}$ scale under (R); primitive sufficient conditions, inherited from the rate conditions behind Arkhangelsky et al. (2021)'s Theorem 1, are discussed in the supplementary appendix, which also characterizes when (R) is plausible. In finite samples, the bootstrap-based intervals we recommend capture the contributions of (ii)–(iii) regardless of whether (R) holds exactly. □

Proposition 3 (Cluster-robust variance). *Suppose treatment is assigned at the cluster level, with* K *clusters indexed* $k = 1, \ldots, K$, $K \to \infty$ *and* $\max_k \tilde{\omega}^{(k)} \to 0$ *where* $\tilde{\omega}^{(k)} = \sum_{j \in k} \tilde{\omega}_j$ *is the cluster-aggregated weight, and suppose errors are independent across clusters but arbitrarily dependent within them. Then the cluster bootstrap variance estimator*

$$\widehat{V}_w^{cb} = \frac{1}{B-1} \sum_{b=1}^{B} \left(\hat{\tau}^{w,(b)} - \bar{\tau}^w\right)^2$$

is consistent for the cluster-robust asymptotic variance—which augments σ_ξ^2 with within-cluster covariance terms—where $\hat{\tau}^{w,(b)}$ is computed from a cluster-resampled bootstrap draw with weight renormalization $\tilde{\omega}_j^{(b)} = \tilde{\omega}_j n_j^{(b)} / \sum_{\ell=1}^{N_1} \tilde{\omega}_\ell n_\ell^{(b)}$.*

Proof sketch. The central limit argument of Proposition 2 is applied one level up, to cluster-aggregated noise contrasts whose variances absorb within-cluster covariances; standard cluster-bootstrap arguments (Cameron et al., 2008; Clarke et al., 2023) then apply to the bilinear form (8) provided the cluster-level weights satisfy the no-dominant-cluster condition—the Lindeberg analogue at the cluster level. The renormalization step preserves the convex-combination structure of $\tilde{\omega}$ within each draw. No bound on cluster sizes is required; what matters is that no cluster's weight share dominates. □

The companion supplementary appendix gives full proofs of all three propositions. Our purpose here is to characterize the regime in which inference based on (8) has the standard frequentist interpretation.

When the asymptotics lose their bite. Condition (W2)—and its cluster-level analogue in Proposition 3—is not innocuous in applications. When researchers use population weights in panels with one or two dominant units (e.g., California Proposition 99, where one large state is treated; or our ACA application, where Los Angeles County alone carries roughly 6% of the population-weighted treated mass), the effective sample size cannot exceed the inverse squared share of the largest unit and in practice sits far below N_1, however many small units the panel adds. A sequence of such weights may still satisfy (W2) formally, but the convergence is too slow for the asymptotic approximation to be reliable at realistic sample sizes. In these settings, plug-in asymptotic standard errors will understate uncertainty, and one must rely on either (i) cluster-robust bootstrap inference at a level of aggregation where Proposition 3's no-dominant-cluster condition is plausible, or (ii) randomization-inference / placebo-distribution tests that condition on the actual weight structure. We pursue both routes in our application.

2.4 Inference

We adapt the three variance estimators of Arkhangelsky et al. (2021) (bootstrap, jackknife, and placebo) to the weighted case. The key complication in resampling is the renormalization of $\tilde{\omega}$: when treated units are dropped or duplicated, the weights must be renormalized to remain a probability vector.

Bootstrap. In bootstrap draw b, treated unit j appears $n_j^{(b)}$ times, and the bootstrap-adjusted weights are $\tilde{\omega}_j^{(b)} = \tilde{\omega}_j n_j^{(b)} / \sum_k \tilde{\omega}_k n_k^{(b)}$. Control weights $\hat{\omega}$ are similarly renormalized. The cluster bootstrap of Clarke et al. (2023) generalizes this to the case where clusters of units are resampled together; we extend the renormalization to act at both unit and cluster levels.

Jackknife. The leave-one-out jackknife renormalizes the remaining weights when a treated unit is dropped: $\tilde{\omega}_k^{(-j)} = \tilde{\omega}_k / \sum_{\ell \neq j} \tilde{\omega}_\ell$ for $k \neq j$. We use the *fixed-weight* jackknife, holding $\hat{\omega}^w$ and $\hat{\lambda}^w$ at their full-sample values. Following Arkhangelsky et al. (2021), the jackknife is not recommended for the SC variant due to non-differentiability at the boundary of the simplex.

Placebo. The placebo method assigns N_1 control units as pseudo-treated and estimates the treatment effect on them. Since the original $\tilde{\omega}$ do not naturally map to control units, our default uses uniform weights for pseudo-treated units, with software options for size-matched and permuted weighting schemes. The cluster-robust placebo extension is left to future work.

Diagnosing estimand divergence. Because $\hat{\tau}^w$ and $\hat{\tau}^{\text{sdid}}$ estimate different estimands, we recommend that practitioners estimate both and test whether the gap between them exceeds sampling noise. The paired bootstrap does this directly: apply identical unit (or cluster) resamples to both estimators in each replication, compute $\Delta_b = \hat{\tau}_b^w - \hat{\tau}_b$, and compare the observed gap to the recentered bootstrap distribution of Δ_b. This is a Hausman-style comparison. A significant gap indicates treatment-effect heterogeneity correlated with the weights, so the estimand choice of Section 2.2 is consequential and should be made—and defended—explicitly; an insignificant gap means the weighting choice is innocuous and the equally-weighted default loses nothing. We view this as the natural ex-ante diagnostic for whether weighted SDID is needed in a given application, and we apply it in Section 3.2.

The supplementary appendix states each estimator in full; Section 4 reports a Monte Carlo evaluation of finite-sample bias and coverage.

3 Application: The ACA Medicaid Expansion

We apply the weighted SDID estimator to the county-level mortality effects of the ACA's 2014 Medicaid expansion, building on the design of Borgschulte and Vogler (2020). The ACA application is well-suited to demonstrating the contribution of the weighted estimator: counties differ in population by three orders of magnitude, treatment was assigned at the state level (so within-state correlation matters for inference), and there is plausible heterogeneity in mortality response across the urban–rural gradient given documented differences in baseline uninsurance rates (Sommers et al., 2016; Miller et al., 2021).

3.1 Setting and Data

The sample is a balanced county-year panel covering 2009–2017 with 2,824 counties (1,181 treated, 1,643 control). The outcome is the all-cause mortality rate per 100,000 among adults aged 20–64, drawn from CDC WONDER's Underlying Cause of Death file (Centers for Disease Control and Prevention, National Center for Health Statistics, 2021). We use these public-use tabulations rather than the restricted-access National Vital Statistics System microdata used by Borgschulte and Vogler (2020); both derive from the same NCHS underlying records. The treated counties are those in the 26 jurisdictions (25 states and the District of Columbia) that expanded Medicaid coverage to childless adults effective January 1, 2014 under the ACA. Following Borgschulte and Vogler (2020), the six states that expanded *after* January 2014 (Alaska, Indiana, Louisiana, Montana, New Hampshire, and Pennsylvania) are excluded from the panel entirely: over the sample period they are neither cleanly treated nor valid never-treated controls. The remaining 45 jurisdictions (44 states plus the District of Columbia) constitute the clusters for state-level inference. Treated-unit weights $\tilde{\omega}_j$ are each county's share of total treated-county population as of 2013. The population distribution is highly skewed: the 90th percentile county (185,562 residents) is an order of magnitude larger than the 10th percentile (3,845).

The skewness of the weight distribution is consequential for inference. The Herfindahl index of the population weights is $H = 0.0098$, implying an effective number of treated units $N_1^{\text{eff}} = 102$ relative to the raw $N_1 = 1181$. A single county—Los Angeles County—accounts for roughly 6.1% of total population-weighted treated mass; the ten largest counties account for 22%. This concentration leaves the effective-sample-size condition (W2) of Proposition 1 with little practical bite, so we rely on cluster-robust bootstrap inference at the state level, where treatment is assigned. The no-dominant-cluster condition of Proposition 3 is more plausible across the 26 treated state clusters, though not comfortably satisfied—California alone accounts for roughly 23% of treated population mass (Supplementary Appendix B.5)—which is why we complement the cluster-bootstrap intervals with randomization inference via the in-space placebo (Section 3.6).

Spillovers and SUTVA. Identification treats non-expansion counties as unaffected by neighboring states' expansions. Cross-border spillovers could violate this: newly insured patients can use providers across state lines, hospital systems near borders serve mixed patient pools, and expansion may have induced enrollment among already-eligible individuals elsewhere (Frean et al., 2017; Sommers et al., 2016). Two considerations limit the threat to our conclusions. First, the plausible direction of health-improving spillovers onto control counties *attenuates* the estimated treated–control contrast; spillovers of this kind cannot manufacture the negative population-weighted estimate, which would if anything be larger in their absence. Second, our central result—the divergence between the equally-weighted and population-weighted estimands—is a comparison within the treated group: both estimands rest on the same control pool and the same estimator class, so contamination of control counties moves the two estimates together rather than apart. Spillovers *within* treated states (say, from urban treatment intensity to neighboring rural counties) would color the interpretation of the size gradient in Figure 2, but not the existence of the divergence.

3.2 Main Results

Table 1: Effect of the 2014 Medicaid expansion on county-level mortality.

		Equally weighted			Population weighted		
Estimator	Covariates	Estimate	SE (boot)	SE (cluster)	Estimate	SE (boot)	SE (cluster)
DID	No	-0.13	2.26	3.92	-18.39***	2.25	4.75
DID	Yes	-0.67	2.17	3.97	-19.52***	2.46	5.80
SDID	No	-0.71	2.42	3.45	-17.45***	2.21	4.66
SDID	Yes	-1.18	2.20	3.65	-18.66***	2.51	5.16

Notes:
Outcome: all-cause mortality per 100,000, ages 20–64 (CDC WONDER). Sample: 2,824 counties, 2009–2017. Each cell reports the same estimator under two standard errors: SE (boot) is the county-level bootstrap; SE (cluster) is the state-clustered bootstrap (45 states), both with 500 replications. Covariate rows adjust for the six time-varying county controls of Borgschulte and Vogler (2020): percent white, percent aged 55–64, log population aged 20–64, log population aged 35–44, log female population aged 20–64, and the unemployment rate; covariates for three FIPS-change counties (15005, 46113, 51019) are imputed with state-year means so all rows share the same sample. Significance stars use the state-clustered bootstrap SE: * p<0.10, ** p<0.05, *** p<0.01. Synthetic control results are reported and diagnosed in Supplementary Appendix E.1.

Table 1 reports the main results. Across both estimators, with and without covariate adjustment, the equally-weighted and population-weighted estimates diverge sharply: the equally-weighted estimates are statistically indistinguishable from zero while their population-weighted counterparts are large and negative. The equally-weighted SDID estimate is -0.71 deaths per 100,000 (state-clustered bootstrap SE 3.45); the population-weighted SDID estimate is -17.45 (SE 4.66), significant at the 1% level. The DID estimates show the same pattern. Adjusting for the six time-varying county covariates used by Borgschulte and Vogler (2020) leaves both conclusions essentially unchanged (-1.18 equally weighted, -18.66 population weighted for SDID), confirming that the divergence is a property of the weighting choice rather than of omitted time-varying confounders. For every specification we report both the county-level bootstrap and the state-clustered bootstrap standard error; clustering at the state level—the level at which treatment is assigned—increases the standard errors but does not change any qualitative conclusion. We exclude synthetic control from Table 1: its estimates are an extreme outlier relative to the DID–SDID family in this application, and we report them, together with the standard battery of post-estimation diagnostics that explains their magnitude, in Supplementary Appendix E.1.

To confirm that the divergence between the equally-weighted and population-weighted SDID estimates is not attributable to sampling variability, we apply the paired-bootstrap divergence diagnostic recommended in Section 2.4. On each of $B = 500$ bootstrap replications we apply identical unit resamples to both estimators and compute $\Delta_b = \hat{\tau}_b^w - \hat{\tau}_b$. Recentering the bootstrap distribution to the null of no difference, the two-sided p-value for the observed gap of $|\Delta_{\text{obs}}| = 16.7$ deaths per 100,000 is $p < 0.001$. The two estimands are sharply distinguishable—in the recommended workflow, exactly the situation in which the researcher must take an explicit stand on the target estimand before reading off a policy conclusion.

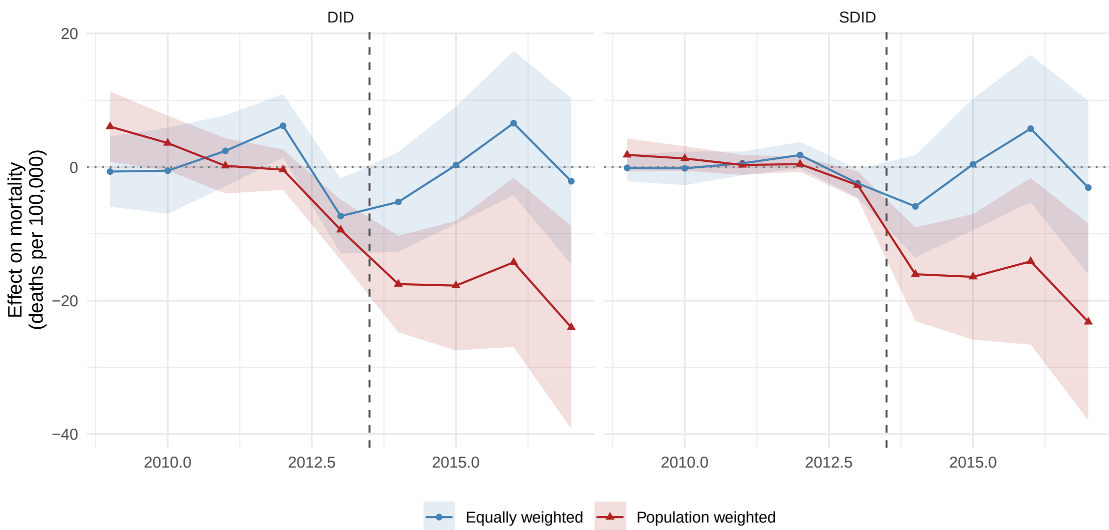

Figure 1: Event-study decomposition under equal and population weighting.
Notes: DID (left) and SDID (right). Points are year-specific effects under equal (blue circles) and population (red triangles) weighting of treated counties; shaded bands are 95% state-clustered bootstrap confidence intervals (500 replications). The vertical dashed line marks treatment onset (2014).

Figure 1 displays the event-study decomposition. Under equal weighting (blue), pre-treatment coefficients cluster near zero for both estimators. Under population weighting (red), the DID panel shows non-trivial pre-treatment divergence between treated and control groups, raising parallel-trends concerns when larger counties are weighted more heavily—precisely the concern that motivated Borgschulte and Vogler (2020)'s DLPS approach. The SDID estimator addresses the pre-treatment fit problem through a different route: its data-driven time weights $\hat{\lambda}^w$ down-weight periods in which treated and synthetic-control trajectories diverge, yielding tighter pre-treatment coefficients in the right panel (RMSPE: 1.38 for SDID vs. 4.44 for DID, both equally weighted). Post-treatment, both estimators show the same qualitative pattern: under population weighting the coefficients are consistently negative (-11 to -17 deaths per 100,000); under equal weighting they fluctuate near zero. The consistency across estimators isolates the *weighting choice*, not the *estimator choice*, as the source of the divergence.

3.3 Mechanism: Heterogeneity by County Size

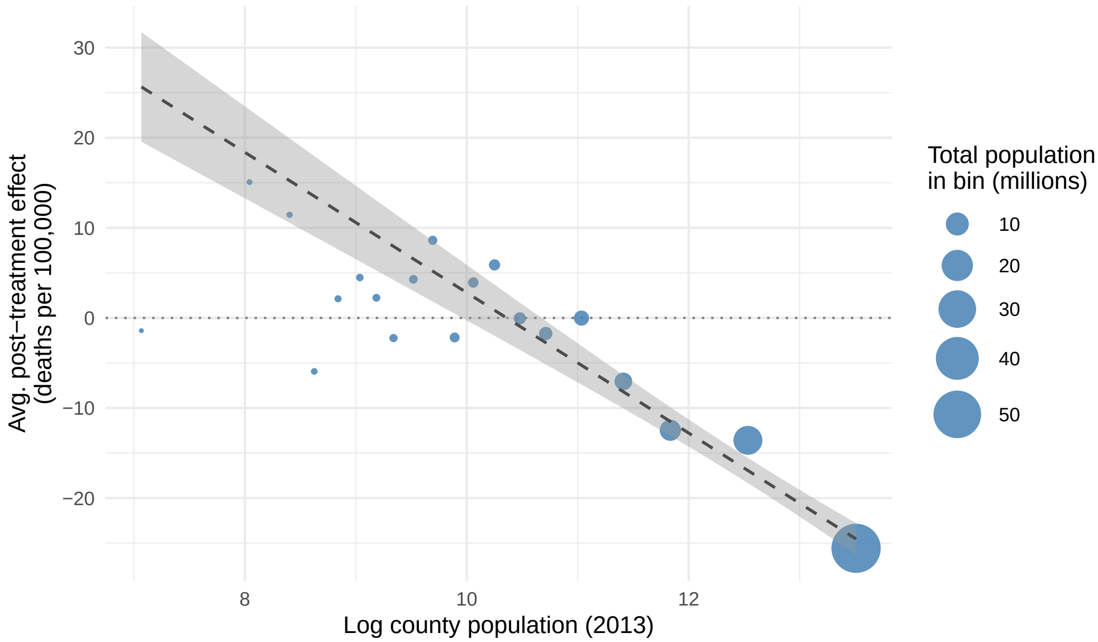

Figure 2: Treatment-effect heterogeneity by county population.
Notes: Binned scatter of unit-level SDID treatment effects against log county population (2013), in 20 quantile bins of log population. Point size is proportional to the total 2013 population of counties in the bin (so larger-population deciles render as larger points even though each bin holds the same number of counties). Unit-level effects use the equally-weighted SDID control weights $\hat{\omega}$ and time weights $\hat{\lambda}$ (the post-period gap relative to the synthetic control minus the $\hat{\lambda}$-weighted pre-period gap), isolating heterogeneity in the treatment effect from heterogeneity in the weighting scheme; the equally-weighted average of the unit-level effects recovers the SDID estimate exactly. The dashed line is a population-weighted OLS fit through the bin means.

The divergence between estimands is not a statistical artifact: it reflects substantive treatment-effect heterogeneity that is correlated with county size. Figure 2 plots mean SDID unit-level effects against log county population in 20 quantile bins of log population, with point area scaled to the total 2013 population in each bin so the visual emphasis matches the population-weighted estimand. Unit-level effects are computed with the *equally-weighted* SDID weights—the post-period gap between each treated county and the synthetic control $\hat{\omega}^\top Y_{\text{control},t}$, net of the $\hat{\lambda}$-weighted pre-period gap—so the heterogeneity displayed is in the treatment effect itself, not in the weighting scheme, and the equally-weighted average of the unit-level effects reproduces the SDID estimate by construction. Across the bottom three population quartiles, mean effects oscillate around zero (between -6 and $+15$ deaths per 100,000) with no systematic pattern. The mortality response is concentrated in the top population bins: mean effects decline monotonically over the four largest-population bins, reaching -26 deaths per 100,000 in the largest bin (log population ≈ 13.5). The population-weighted bin-level correlation between mean log population and mean treatment effect is -0.96; the unweighted county-level correlation is more modest (-0.14), reflecting noise in county-level synthetic control predictions for small counties where mortality rates are volatile. As an internal-consistency check, the *population-weighted* average of these same unit-level effects is -17.8 deaths per 100,000—nearly identical to the weighted SDID estimate of -17.5, even though the unit-level effects hold $\hat{\omega}$ and $\hat{\lambda}$ at their equally-weighted values. The divergence between the two estimands is driven by *which counties* the estimand averages over, not by the re-optimization of the synthetic control.

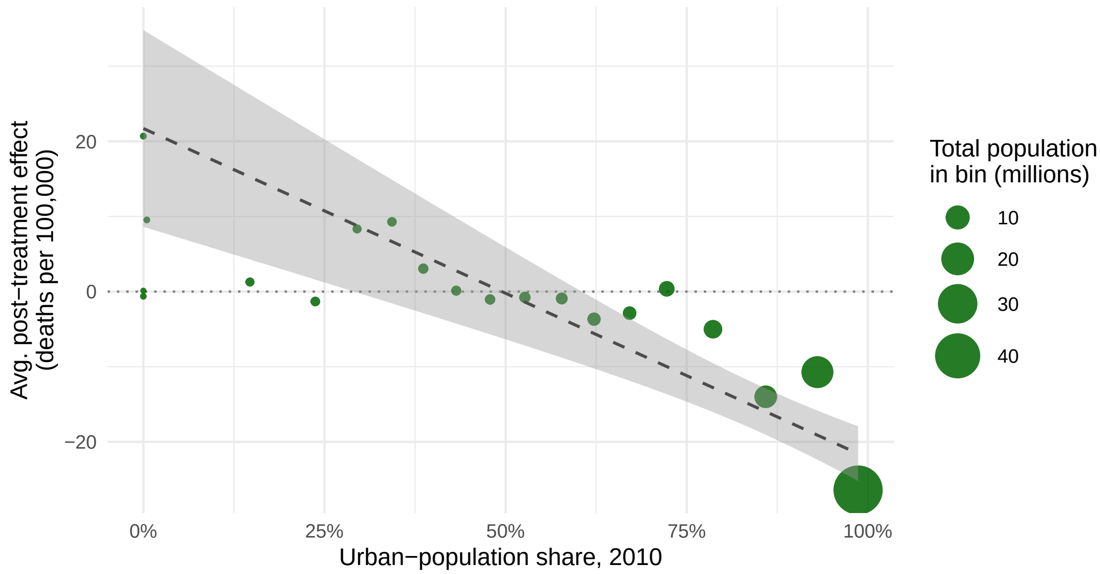

Figure 3: Treatment-effect heterogeneity by urban-population share.
Notes: Binned scatter of unit-level SDID treatment effects against the urban-population share of each county, in 20 quantile bins of pct. urban (2010 Decennial Census, P002002 / P002001). Point area is proportional to the total 2013 population of counties in the bin; the dashed line is a population-weighted OLS fit through the bin means. Urban share is fixed at its 2010 (pre-treatment) value, so the gradient reflects pre-existing county composition rather than any post-2014 reclassification.

Figure 3 repeats the exercise using each county's 2010 urban-population share, fixed at its pre-treatment value to avoid contamination from any post-2014 reclassification. The pattern is the same as in Figure 2 but maps onto a more directly interpretable substantive dimension: the most-rural bin of counties (the bottom twentieth by urban share, $\approx 0\%$ urban) shows mean effects of +21 deaths per 100,000—a noisy positive point estimate, the urban-share counterpart of the around-zero oscillation among small-county bins in Figure 2, not a precisely measured harm—while the most-urban bin ($\approx$ 99% urban) shows reductions of 26. The population-weighted bin-level correlation is -0.83 (unweighted county-level: -0.13). Urban share and log population are mechanically correlated, so this view does not provide an independent test, but it makes the substantive content of the heterogeneity gradient more transparent: the population-weighted estimand is dominated by counties that are both populous *and* highly urbanized—precisely the counties for which Sommers et al. (2016) and Miller et al. (2021) document larger pre-expansion uninsurance rates and more health-care infrastructure capable of absorbing the coverage shock.

Whatever its substantive interpretation, the gradient mechanically generates the divergence between the equally-weighted and population-weighted estimands: equal weighting averages over the many small counties whose estimated effects oscillate around zero; population weighting shifts the estimand toward the few large counties where effects are large and negative.

3.4 Robustness to Alternative Weight Specifications

Table 2: Robustness to alternative treated-unit weight specifications.

Weight specification	Estimate	SE	N_1^{eff}
Population (2013, baseline)	-17.45	4.66	102
Population (2010 base year)	-17.16	4.99	104
Square root of population	-8.11	4.34	569
Log population	-1.84	3.76	1153
Equal ($1/N_1$)	-0.71	3.45	1181

Notes:
SDID estimates with state-clustered bootstrap SEs (500 replications) for every row, including equal weighting. N_1^{eff} is the Kish effective sample size $(\sum_j \tilde{\omega}_j^2)^{-1}$. The baseline weights are each county's share of total treated-county population in 2013; alternative rows change the base year or compress the weight distribution toward equality.

Table 2 examines the sensitivity of the population-weighted SDID estimate to alternative weight specifications. Replacing 2013 population with 2010 population shares yields an estimate of similar magnitude (-17.2 deaths per 100,000), confirming that the result does not depend on the choice of base year. As the weight distribution is compressed—square-root population, log population, equal weighting—the estimate moves monotonically toward zero. The effective sample size grows correspondingly, from $N_1^{\text{eff}} \approx 102$ under linear population weights to $N_1 = 1181$ under equal weights. The substantive lesson is that the large negative effect requires giving substantial weight to the largest counties: log compression alone is sufficient to eliminate it. This is not an artifact of the specific 2013 linear-population specification, but it does pinpoint *which* aggregation choices produce the result.

Table 3: Comparison with the published estimate.

Method	Estimate	SE
Standard SDID (equally weighted)	-0.71	3.45
Weighted SDID (population)	-17.45	4.66
Weighted DID (population)	-18.39	4.75
Borgschulte & Vogler (2020), Table 2 col. 2	-11.36	3.59

Notes:
The first three rows are estimated on our balanced county-year panel with state-clustered bootstrap SEs; the fourth row is the headline DLPS-weighted DID estimate published by Borgschulte and Vogler (2020) (Table 2, column 2; state-clustered SE, 45 clusters). Supplementary Appendix E.5 documents our public-data DLPS reproduction, which recovers the structure of the BV results, including their heterogeneity by baseline uninsurance, on a near-identical county sample.

3.5 Comparison with the Published Estimate

Table 3 places our population-weighted SDID estimate alongside the headline estimate from Borgschulte and Vogler (2020), who weight a propensity-score-adjusted DID by working-age county population using a double-lasso variable selection procedure on a rich set of demographic, economic, and political characteristics. Our population-weighted SDID estimate (-17.45 deaths per 100,000) is in the same neighborhood as BV's published -11.36, despite differences in identification strategy (data-driven control matching plus pre-treatment time-weight selection vs. a propensity score for treatment selection) and in data source (CDC WONDER public tabulations vs. restricted NVSS microdata). The corresponding equally-weighted SDID estimate is statistically indistinguishable from zero, underscoring that the choice of treated-unit weights—not the choice of estimator—accounts for nearly all of the gap. Supplementary Appendix E.5 reports our public-data reproduction of the BV design itself: recovering CDC WONDER's suppressed small-count cells by allocating exact state-year residuals brings the estimation sample within two counties of BV's, and the reproduction recovers their qualitative results, including the concentration of the mortality response in counties with high baseline uninsurance.

3.6 Placebo Tests

We conduct two complementary falsification exercises. The *in-time* placebo restricts the sample to pre-treatment years (2009–2013) and assigns simulated treatment dates of 2011 and 2012. Absent a policy change, the estimator should return zero. Figure 4 (left) shows that the *equally-weighted*

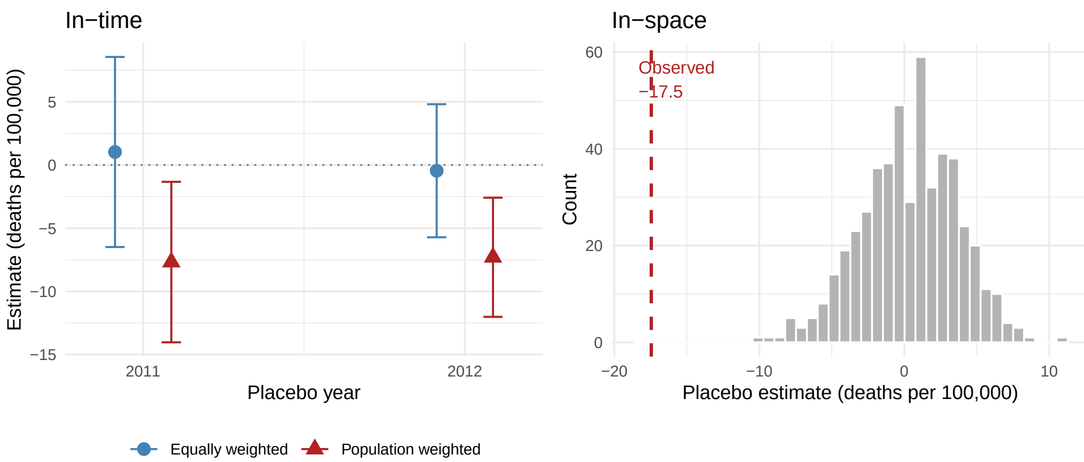

Figure 4: Placebo tests.
Notes: Left: in-time placebo with simulated treatment dates of 2011 and 2012, restricting to pre-treatment data (2009–2013); circles are equally-weighted and triangles population-weighted SDID estimates, with whiskers marking 95% state-clustered bootstrap confidence intervals (500 replications). Right: in-space placebo distribution from 500 random re-assignments of treatment to control counties, with the dashed vertical line marking the observed population-weighted estimate.

placebo estimates are statistically indistinguishable from zero. The *population-weighted* estimates, however, are modestly negative (-7.7 and -7.3 deaths per 100,000) with 95% state-clustered bootstrap intervals that exclude zero—the in-time signature of the same pre-existing differential mortality trend in large counties visible in the population-weighted event study (Figure 1). When the estimand is dominated by large urban counties already on a declining trajectory, a placebo onset date partly captures that trend. The pre-treatment divergence is markedly smaller under SDID than under DID—SDID's data-driven time weights down-weight the divergent pre-periods (Figure 1)—and the residual is precisely why we base inference on the cluster-robust bootstrap of Section 2.4 rather than plug-in asymptotics. The *in-space* placebo randomly re-assigns treatment to 1181 control counties (matching the true number of treated counties) while keeping the real treatment date of 2014 and using equal weights. Figure 4 (right) shows the resulting distribution of placebo estimates from 500 random draws; the observed population-weighted estimate (-17.5) sits well in the left tail, yielding a randomization-inference p-value of 0.

4 Monte Carlo Evidence

We evaluate the finite-sample performance of the weighted SDID estimator and its variance estimators via simulation. The key questions are: (i) is the weighted estimator unbiased for its target $\tau^{(\tilde{\omega})}$ under heterogeneous treatment effects? (ii) do the bootstrap, jackknife, and placebo variance estimators achieve nominal coverage when treatment effects are correlated with unit size?

4.1 Design

The data-generating process follows the factor model of Arkhangelsky et al. (2021). For $N = N_0 + N_1$ units observed over $T = T_0 + T_1$ periods,

$$Y_{it} = L_{it} + \tau_{it} W_{it} + \sigma \epsilon_{it},$$

where $L = UV^\top + \alpha \mathbf{1}^\top + \mathbf{1}\beta^\top$ is rank-2 with additive fixed effects, $U \in \mathbb{R}^{N\times 2}$ and $V \in \mathbb{R}^{T\times 2}$ are factor loadings drawn from Poisson distributions, $\alpha_i = 10i/N$, $\beta_t = 10t/T$, and ϵ_{it} is drawn from an AR(1) process with $\rho = 0.7$. Each treated unit j is assigned a "size" s_j from a log-normal distribution scaled to achieve a target Herfindahl index for the population weights $\tilde{\omega}_j = s_j / \sum_k s_k$, and the treatment effect is

$$\tau_{ij} = \bar{\tau} + \gamma \left(\frac{s_j}{\bar{s}} - 1 \right),$$

with $\bar{\tau} = 1$, $\bar{s} = N_1^{-1} \sum_j s_j$, and $\gamma \geq 0$ controlling the size-effect correlation. We fix $N_0 = 100$, $T_0 = 40$, $T_1 = 10$, $\sigma = 0.5$, and vary $N_1 \in \{5, 10, 20\}$, $\gamma \in \{0, 0.5, 1.0\}$ (homogeneous, moderate, strong heterogeneity), and the weight concentration (low, HHI $= 1.5/N_1$; high, HHI $= 3/N_1$), yielding 18 grid configurations evaluated over 100 simulations each with bootstrap and placebo variance estimators using 50 replications. A nineteenth, ACA-calibrated configuration gives the application's standard errors direct simulation backing: it reproduces the application's panel dimensions ($N_0 =$ 1,643, $N_1 =$ 1,181, $T_0 = 5$, $T_1 = 4$) and sets the treated-unit sizes to the application's actual 2013 county populations, so every simulation has exactly the application's weight concentration (HHI $= 0.0098$, $N_1^{\text{eff}} \approx 102$), with $\gamma = 1$, the strong size-effect-correlation regime the application's county-size gradient suggests. The simulation sweep is parallelized across configurations and simulation draws (`scripts/run_mc_simulations.R`), with seeding that reproduces the equivalent sequential loop exactly; implementation details appear in Supplementary Appendix D.

4.2 Results

Tables 4 and 5 report bias, RMSE, and 95% confidence-interval coverage across the simulation configurations. Three patterns are central to the inference recommendations of Section 2:

- Both the equally-weighted and population-weighted SDID estimators are approximately unbiased for their respective targets, with RMSE differences favoring the equally-weighted variant for the obvious reason that it has a larger effective sample.

- The bootstrap and placebo variance estimators achieve approximately nominal 95% coverage, with undercoverage at high HHI for the placebo (which uses uniform pseudo-treated weights, as discussed in Section 2.4).
- The jackknife is near-nominal with few treated units and grows conservative as the number of treated units increases, over-covering at $N_1 = 20$.

These patterns validate Propositions 1–3 in finite samples within the regime where Assumptions W2–W3 are quantitatively reasonable.

Table 4: Monte Carlo bias and RMSE of the SDID estimators.

			Equally weighted		Population weighted	
N_1	γ	HHI	Bias	RMSE	Bias	RMSE
5	0.0	0.3000	0.022	0.188	0.002	0.223
5	0.0	0.6000	0.012	0.177	-0.002	0.305
5	0.5	0.3000	0.012	0.177	0.006	0.224
5	0.5	0.6000	0.012	0.177	-0.002	0.305
5	1.0	0.3000	0.012	0.177	0.006	0.224
5	1.0	0.6000	0.012	0.177	-0.002	0.305
10	0.0	0.1500	0.001	0.117	0.004	0.148
10	0.0	0.3000	-0.013	0.129	-0.013	0.217
10	0.5	0.1500	-0.013	0.129	-0.013	0.163
10	0.5	0.3000	-0.013	0.129	-0.013	0.217
10	1.0	0.1500	-0.013	0.129	-0.013	0.163
10	1.0	0.3000	-0.013	0.129	-0.013	0.217
20	0.0	0.0750	0.004	0.075	0.004	0.083
20	0.0	0.1500	0.004	0.076	0.012	0.121
20	0.5	0.0750	0.004	0.076	0.007	0.090
20	0.5	0.1500	0.004	0.076	0.012	0.121
20	1.0	0.0750	0.004	0.076	0.007	0.090
20	1.0	0.1500	0.004	0.076	0.012	0.121
1181	1.0	0.0098	-0.001	0.023	0.005	0.053

Notes:
Bias and RMSE are computed against each estimator's own target ATT (the equally-weighted and population-weighted averages of the unit-level effects, respectively). 100 simulations per configuration. N_1 is the number of treated units, γ the size-effect correlation, and HHI the Herfindahl index of the treated-unit weights. Data-generating process described in Section 4.1.

Table 5: Monte Carlo 95% confidence-interval coverage.

			Equally weighted			Population weighted		
N_1	γ	HHI	Boot	JK	Plac	Boot	JK	Plac
5	0.0	0.3000	0.89	0.92	0.90	0.85	0.85	0.83
5	0.0	0.6000	0.88	0.86	0.94	0.87	0.87	0.69
5	0.5	0.3000	0.93	0.94	0.94	0.96	0.92	0.83
5	0.5	0.6000	1.00	1.00	0.94	0.99	0.97	0.69
5	1.0	0.3000	0.99	1.00	0.94	1.00	1.00	0.83
5	1.0	0.6000	1.00	1.00	0.94	1.00	0.99	0.69
10	0.0	0.1500	0.97	0.96	0.94	0.92	0.91	0.88
10	0.0	0.3000	0.90	0.88	0.90	0.84	0.84	0.74
10	0.5	0.1500	0.99	0.99	0.90	0.95	0.94	0.83
10	0.5	0.3000	1.00	1.00	0.90	0.97	0.98	0.74
10	1.0	0.1500	1.00	1.00	0.90	0.99	0.99	0.83
10	1.0	0.3000	1.00	1.00	0.90	1.00	1.00	0.74
20	0.0	0.0750	0.97	0.97	0.97	0.95	0.95	0.96
20	0.0	0.1500	0.96	0.97	0.97	0.91	0.92	0.82
20	0.5	0.0750	0.99	0.99	0.97	1.00	1.00	0.96
20	0.5	0.1500	1.00	1.00	0.97	1.00	1.00	0.82
20	1.0	0.0750	1.00	1.00	0.97	1.00	1.00	0.96
20	1.0	0.1500	1.00	1.00	0.97	1.00	1.00	0.82
1181	1.0	0.0098	1.00	1.00	1.00	1.00	1.00	0.74

Notes:
Nominal level 0.95. Bootstrap (Boot) and placebo (Plac) variance estimators use 50 replications; JK is the fixed-weight jackknife. 100 simulations per configuration. Coverage is evaluated against each estimator's own target ATT.

The ACA-calibrated configuration (the $N_1 = 1{,}181$ rows of Tables 4 and 5) speaks directly to the application: at the application's own panel dimensions and weight concentration, the population-weighted estimator has bias 0.005 against its target (RMSE 0.053, target ATT 11.6), and the bootstrap attains 100% coverage, the jackknife 100%, and the uniform-weight placebo 74%. The bootstrap's performance in this cell is the simulation counterpart of the cluster-robust bootstrap we rely on for Table 1.

5 Discussion

We have shown that the SDID estimator of Arkhangelsky et al. (2021) admits a parsimonious extension to incorporate researcher-specified weights on treated units, with consistency and asymptotic normality results that inherit from the underlying factor-model framework subject to a no-dominant-weight regularity condition. We have also documented—through both formal characterization and direct application—that this regularity condition loses its practical bite in many policy settings, particularly those involving population weights on heterogeneously sized geographic units. In such settings, plug-in asymptotic standard errors are unreliable; cluster-robust bootstrap inference at a level of aggregation where the no-dominant-cluster condition is plausible is the appropriate finite-sample tool. Our application provides a concrete illustration: with $N_1^{\mathrm{eff}} \approx 102$ and one county carrying $\approx 6.1\%$ of total weight, the asymptotic approximation cannot be trusted at face value, but state-level cluster bootstrapping across the 26 treated state clusters—complemented by randomization inference—provides a defensible alternative.

The ACA application illustrates more than the technical contribution. The equally-weighted and population-weighted estimands diverge by 17 deaths per 100,000—enough to swing the qualitative interpretation of the policy from "no detectable effect" to "substantial mortality reduction." The choice between these estimands is not a statistical matter but a substantive one. A researcher interested in the average effect across counties as administrative units might reasonably prefer equal weighting; a researcher interested in the per-capita effect on the population exposed to expanded Medicaid coverage—as is the natural reading for cost-benefit and welfare calculations—should prefer population weighting. Both estimands are well-defined and identifiable; the published mortality literature has implicitly favored the population-weighted target through cluster sampling, propensity-score weighting, or other means (Borgschulte and Vogler, 2020; Miller et al., 2021). Our contribution makes this choice transparent and provides a tool that respects it directly.

Three caveats merit discussion. First, our treated-unit weights are fixed by the researcher rather than estimated. We do not claim optimality in any decision-theoretic sense; the weights encode the researcher's definition of the estimand. Adapting the weights to optimally estimate a target subpopulation effect (in the spirit of Hirano et al. (2003) for propensity-score weighting) is a natural extension we leave for future work. Second, while we provide cluster-robust bootstrap and jackknife inference, our adaptation of the placebo variance estimator uses uniform weights for pseudo-treated units, which may understate variance when treatment effects are highly heterogeneous; the implications for finite-sample coverage are examined in the companion supplement. Third, our framework assumes simultaneous adoption of treatment. Extension to staggered settings, where the weighted estimator can be applied within each adoption cohort and aggregated with researcher-chosen cohort weights, raises additional questions about the interaction between within-cohort and across-cohort weighting that are left to future work—and which connect naturally to the broader literature on aggregation in heterogeneous-effects panel designs (Callaway and Sant'Anna, 2021;

Sun and Abraham, 2021; de Chaisemartin and D'Haultfoeuille, 2020).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude Code (Anthropic) for R package development, code development, data processing, replication of prior work, code review, and manuscript editing. The authors reviewed and verified all analyses, results, and text, and take full responsibility for the content of the published article.

Data availability

Data used in this study are publicly available from CDC WONDER (mortality, Underlying Cause of Death file, all-cause mortality among adults aged 20–64, ICD-10 codes A00–Y89; county-year cells with 1–9 deaths are suppressed in the public file and are imputed by allocating exact state-year residuals, as described in Supplementary Appendix E.5) (Centers for Disease Control and Prevention, National Center for Health Statistics, 2021); the SEER program (county-level population by age and sex) (National Cancer Institute, Surveillance Research Program, 2021); and the Bureau of Labor Statistics (county-level unemployment, Local Area Unemployment Statistics) (U.S. Bureau of Labor Statistics, 2025). The BV reproduction in Supplementary Appendix E.5 additionally uses the ACS 2009–2013 5-year file and Census SAHIE 2013 (via the Census API), and county presidential returns from the MIT Election Data and Science Lab. Replication code, the weighted `synthdid` R package extension, and a companion supplementary appendix containing full proofs of Propositions 1–3, synthetic-control diagnostics, alternative weight specifications, leave-one-out sensitivity, and a complete description of the DLPS implementation are available at https://github.com/johniselin-econ/synthdid_weights.

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